# LaMont D. Walker

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## Education

Ph.D., Mathematics, Howard University, Washington, DC, expected June 2030

B.A., Mathematics, University of Washington, Seattle, WA, June 2026

### Relevant Coursework

Advanced Linear Algebra, Differential Geometry, Modern Algebra, Topology, Complex Analysis, Real Analysis

### Independent Reading

*Foundations of Algebraic Geometry, Vakil*

- Translate between vanishing sets, their associated ideals and the distinguished open sets of the spectrum of a commutative ring.
- Prove the spectrum is compact using a partition of unity argument
- Show that a morphism of ringed spaces induces a homomorphism of the associated stalks
- Prove the lying over theorem by showing that for any field which is an integral extension of one of its subrings, that subring is also a field.

*Galois Theory, Leinster*

- View the Galois group of a polynomial as the subgroup of the permutations which fix conjugate tuples of the roots.
- Learn about the Galois correspondence, the connection with solvability of groups to solvability of a polynomial by radicals, how to use the Galois correspondence to classify finite fields.

*Algebraic Topology, Hatcher*

- Learn about deformation retractions, homotopy, homotopy equivalence, the fundamental group and the Van Kampen theorem
- Learn the proof of Brouwer's Fixed Point Theorem on a closed disk using the fundamental group of the circle.

*Category Theory in Context, Riehl*

- Learn about functors, natural transformations, limits and colimits, adjunctions and Kan extensions.
- Derive limits, colimits, and adjunctions as special cases of Kan extensions
- Prove the Yoneda Lemma

## Research Experience

### Summer Research in Mathematics at UW

Seattle, WA, Summer 2025

- Gave lecture on Category Theory and Operadic Lie Algebras
- Read Hu's and Venkatesh's papers on Ver+4
- Learned about monoidal categories

### Undergraduate Research Fellowship at UW

Seattle, WA, Summer 2023

- Programmed simulation for the molecular orbitals of a polyacetylene chain using the SSH model
- Applied quantum mechanics and topology to understand the properties of topological insulators
- Gave lectures on the expressions of Maxwell's equations in reciprocal space

## Projects

### Classify All Finite Groups with maximal subgroups of prime order

- Classified the finite groups with maximal subgroups of prime order
- Proved the existence of such groups
- Determined the orders of all groups of that sort
- Determined when such groups were simple, and when they were abelian

## Awards

Early entrance into the University of Washington through the Robinson Center

Rising Chem Star Undergraduate Scholarship

John and Kathy Connors Foundation Endowed Scholarship

## Research Interests

Hilbert schemes of nonreduced algebraic curves

Modular forms, elliptic curves and Galois representations